CM0859 – MT5009 Type Theory
Typed Lambda Calculus (or the Proposition-as-Types Principle)

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Preliminaries

Textbook

Pfenning (2023). Lecture Notes on Proofs as Programs.

Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Source code

The AGDA code have been tested with AGDA 2.8.0.

Introduction

- Propositions (types): $A, B, C, \ldots$

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- Lambda terms (programs): $M, N, \ldots$

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- Propositions (types): $A, B, C, \ldots$
- Lambda terms (programs): $M, N, \ldots$
- Judgements:
  - (i) $A \; \mathrm{type}$ (A is a type)
  - (ii) $M : A$ (λ-term M is a proof-term for proposition A)
    (program M has type A)
  - (iii) $M \Rightarrow_R M'$ (λ-term M reduces to λ-term M')

Introduction

- Proposition-as-types principle

  "We intend that if $M : A$ then $A\,\mathrm{true}$. Conversely, if $A\,\mathrm{true}$ then $M : A$ for some appropriate proof-term M. But we want something more: every deduction of $M : A$ should correspond to a deduction of $A\,\mathrm{true}$ with an identical structure and vice versa." (p. L4.1)

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  "The introduction rules for a logical connective correspond to the ***constructors*** *for the corresponding type. Conversely, the elimination rules correspond to* ***destructors****." (p. L4.2)*

Introduction

- Proposition-as-types principle

  "We intend that if $M : A$ then $A\ \mathrm{true}$. Conversely, if $A\ \mathrm{true}$ then $M : A$ for some appropriate proof-term M. But we want something more: every deduction of $M : A$ should correspond to a deduction of $A\ \mathrm{true}$ with an identical structure and vice versa." (p. L4.1)

  *"The introduction rules for a logical connective correspond to the **constructors** for the corresponding type. Conversely, the elimination rules correspond to **destructors**." (p. L4.2)*

- Form of the inference rules (same than for natural deduction):

$$\frac{J_1 \quad \ldots \quad J_n}{J}\ \text{rule name}$$

where J (conclusion) and $J_1, \ldots, J_n$ (premises or hypotheses) are all judgements.

Introduction

- Inference rules (same than for natural deduction): Formation, introduction and elimination rules.

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- New inference rules: Reduction rules.

> *“If proofs are programs then we need to explain how proofs are to be executed, and which results may be returned by a computation.”*
>
> *“Reduction arises when a destructor is applied to a constructor.”*
>
> *“A proof reduction arises when we apply an elimination rule to the result of an introduction rules.” (p. L4.7)*

Introduction

- Inference rules (same than for natural deduction): Formation, introduction and elimination rules.
- New inference rules: Reduction rules.[†]

> *"If proofs are programs then we need to explain how proofs are to be executed, and which results may be returned by a computation."*
>
> *"Reduction arises when a destructor is applied to a constructor."*
>
> *"A proof reduction arises when we apply an elimination rule to the result of an introduction rules." (p. L4.7)*

[†] Some authors introduce *computation* rules instead of *reduction* rules.

Introduction

Observation

Other names for the proposition-as-types principle are (or should be):

- the Curry-Howard correspondence,
- the Curry-Howard isomorphism (textbook),
- the Brouwer-Heyting-Kolmogorov-Schönfinkel-Curry-Meredith-Kleene-Feys-Gödel-Läuchli-Kreisel-Tait-Lawvere-Howard-de Bruijn-Scott-Martin Löf-Girard-Reynolds-Stenlund-Constable-Coquand-Huet$\cdots$ correspondence (Sørensen and Urzyczyn 2006, p. viii).

Types

Definition

The **types** of typed lambda calculus are defined by the following grammar:

$$
\begin{aligned}
\sigma, \tau ::= {} & \mathbf{b} && \text{(base type)} \\
\mid {} & \sigma \to \tau && \text{(function type)} \\
\mid {} & \sigma \times \tau && \text{(product type)} \\
\mid {} & \sigma + \tau && \text{((disjoint) sum type or coproduct type)} \\
\mid {} & \mathbf{1} && \text{(unit type)} \\
\mid {} & \mathbf{0} && \text{(empty type)}
\end{aligned}
$$

Terms

Definition

The λ-**terms** of our typed lambda calculus are defined by the following grammar:

$$
\begin{array}{rlr}
M, N, P ::= & x & \text{(variable)} \\
| & \lambda x.M & \text{(λ-abstraction)} \\
| & M\ N & \text{(application)} \\
| & \langle M , N \rangle \mid \mathsf{fst}\ M \mid \mathsf{snd}\ M & \text{(pair and projections)} \\
| & \mathsf{inl}\ M \mid \mathsf{inr}\ M \mid \mathsf{case}(M, x.N, y.P) & \text{(injections and case analysis)} \\
| & \langle\rangle & \text{(unit)} \\
| & \mathsf{abort}\ M & \text{(error)}
\end{array}
$$

Inference Rules for Implication (Function Type)

- Formation rule

$$\frac{A\,\text{prop} \qquad B\,\text{prop}}{A \supset B\,\text{prop},}\ \supset\mathsf{F} \qquad\qquad \frac{A\,\text{type} \qquad B\,\text{type}}{A \supset B\,\text{type}.}\ \supset\mathsf{F}$$

Inference Rules for Implication (Function Type)

- Introduction rule

$$\dfrac{\dfrac{}{A\,\mathrm{true}}\,u \\ \vdots \\ B\,\mathrm{true}}{A \supset B\,\mathrm{true},}\supset\!\mathsf{I}^u \qquad \dfrac{\dfrac{}{u : A}\,u \\ \vdots \\ M : B}{\lambda u.M : A \supset B.}\supset\!\mathsf{I}^u$$

Inference Rules for Implication (Function Type)

- Introduction rule

$$\dfrac{\dfrac{}{A\,\mathrm{true}}\,u \\ \vdots \\ B\,\mathrm{true}}{A \supset B\,\mathrm{true},}\supset\mathsf{I}^u \qquad \dfrac{\dfrac{}{u : A}\,u \\ \vdots \\ M : B}{\lambda u.M : A \supset B.}\supset\mathsf{I}^u$$

- Elimination rule

$$\dfrac{A \supset B\,\mathrm{true} \qquad A\,\mathrm{true}}{B\,\mathrm{true},}\supset\mathsf{E} \qquad \dfrac{M : A \supset B \qquad N : A}{M\,N : B.}\supset\mathsf{E}$$

Inference Rules for Implication (Function Type)

- Introduction rule

$$\dfrac{\dfrac{}{A\ \mathrm{true}}\ u \quad \vdots \quad B\ \mathrm{true}}{A \supset B\ \mathrm{true},}\ \supset\mathsf{I}^u \qquad \dfrac{\dfrac{}{u : A}\ u \quad \vdots \quad M : B}{\lambda u.M : A \supset B.}\ \supset\mathsf{I}^u$$

- Elimination rule

$$\dfrac{A \supset B\ \mathrm{true} \qquad A\ \mathrm{true}}{B\ \mathrm{true},}\ \supset\mathsf{E} \qquad \dfrac{M : A \supset B \qquad N : A}{M\,N : B.}\ \supset\mathsf{E}$$

- Reduction rule

$$(\lambda u.M)\,N \Rightarrow_R M\,[\,u := N\,].$$

Inference Rules for Implication (Function Type)

Observation

For annotating a proof (derivation) with proof terms we proceed top-down as illustrated by the following examples.

Inference Rules for Implication (Function Type)

Example

(i) A proof that $A \supset A\,\mathrm{true}$.

$$\dfrac{\dfrac{}{A\,\mathrm{true}}\,u}{A \supset A\,\mathrm{true}}\supset\!\mathsf{I}^u$$

(ii) A proof that $\lambda u.u : A \supset A$.

$$\dfrac{\dfrac{}{u : A}\,u}{\lambda u.u : A \supset A}\supset\!\mathsf{I}^u$$

Inference Rules for Implication (Function Type)

Example

(i) A proof that $A \supset (B \supset A)\ \mathrm{true}$.

$$\dfrac{\dfrac{\dfrac{}{A\ \mathrm{true}}\,u}{B \supset A\ \mathrm{true}}\,{\supset}\mathsf{I}}{A \supset (B \supset A)\ \mathrm{true}}\,{\supset}\mathsf{I}^u$$

(ii) A proof that $\lambda u.\lambda w.u : A \supset (B \supset A)$.

$$\dfrac{\dfrac{\dfrac{}{u : A}\,u}{\lambda w.u : B \supset A}\,{\supset}\mathsf{I}}{\lambda u.\lambda w.u : A \supset (B \supset A)}\,{\supset}\mathsf{I}^u$$

Inference Rules for Conjunction (Product Type)

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \wedge B\,\mathrm{prop},}\ \wedge\mathsf{F} \qquad\qquad \frac{A\,\mathrm{type} \qquad B\,\mathrm{type}}{A \wedge B\,\mathrm{type}.}\ \wedge\mathsf{F}$$

Inference Rules for Conjunction (Product Type)

- Introduction rule

$$\frac{A\,\mathrm{true} \qquad B\,\mathrm{true}}{A \wedge B\,\mathrm{true},}\ \wedge\mathsf{I} \qquad\qquad \frac{M : A \qquad N : B}{\langle M , N \rangle : A \wedge B.}\ \wedge\mathsf{I}$$

Inference Rules for Conjunction (Product Type)

- Introduction rule

$$\frac{A\,\mathrm{true} \qquad B\,\mathrm{true}}{A \wedge B\,\mathrm{true},}\ \wedge\mathsf{I} \qquad\qquad \frac{M : A \qquad N : B}{\langle M , N \rangle : A \wedge B.}\ \wedge\mathsf{I}$$

- Elimination rules

$$\frac{A \wedge B\,\mathrm{true}}{A\,\mathrm{true},}\ \wedge\mathsf{E}_1 \qquad\qquad \frac{M : A \wedge B}{\mathsf{fst}\,M : A,}\ \wedge\mathsf{E}_1$$

$$\frac{A \wedge B\,\mathrm{true}}{B\,\mathrm{true},}\ \wedge\mathsf{E}_2 \qquad\qquad \frac{M : A \wedge B}{\mathsf{snd}\,M : B.}\ \wedge\mathsf{E}_2$$

Inference Rules for Conjunction (Product Type)

- Introduction rule

$$\frac{A\,\mathrm{true} \qquad B\,\mathrm{true}}{A \wedge B\,\mathrm{true},}\ \wedge\mathsf{I} \qquad\qquad \frac{M : A \qquad N : B}{\langle M , N \rangle : A \wedge B.}\ \wedge\mathsf{I}$$

- Elimination rules

$$\frac{A \wedge B\,\mathrm{true}}{A\,\mathrm{true},}\ \wedge\mathsf{E}_1 \qquad\qquad \frac{M : A \wedge B}{\mathsf{fst}\,M : A,}\ \wedge\mathsf{E}_1$$

$$\frac{A \wedge B\,\mathrm{true}}{B\,\mathrm{true},}\ \wedge\mathsf{E}_2 \qquad\qquad \frac{M : A \wedge B}{\mathsf{snd}\,M : B.}\ \wedge\mathsf{E}_2$$

- Reduction rules

$$\mathsf{fst}\,\langle M , N \rangle \Rightarrow_R M,$$
$$\mathsf{snd}\,\langle M , N \rangle \Rightarrow_R N.$$

Inference Rules for Conjunction (Product Type)

Observation

For annotating a proof (derivation) with proof terms we proceed top-down as illustrated by the following examples.

Inference Rules for Conjunction (Product Type)

Example

(i) A proof that $(A \wedge B) \supset (B \wedge A)\,\mathrm{true}$.

$$\dfrac{\dfrac{\dfrac{}{A \wedge B\ \mathrm{true}}\,x}{B\ \mathrm{true}}\,\wedge\mathsf{E}_2 \qquad \dfrac{\dfrac{}{A \wedge B\ \mathrm{true}}\,x}{A\ \mathrm{true}}\,\wedge\mathsf{E}_1}{\dfrac{B \wedge A\ \mathrm{true}}{(A \wedge B) \supset (B \wedge A)\ \mathrm{true}}\,\supset\mathsf{I}^x}\,\wedge\mathsf{I}$$

(ii) A proof that $\lambda x.\,\langle\, \mathsf{snd}\, x\,,\ \mathsf{fst}\, x\,\rangle : (A \wedge B) \supset (B \wedge A)$.

$$\dfrac{\dfrac{\dfrac{\dfrac{}{x : A \wedge B}\,x}{\mathsf{snd}\, x : B}\,\wedge\mathsf{E}_2 \qquad \dfrac{\dfrac{}{x : A \wedge B}\,x}{\mathsf{fst}\, x : A}\,\wedge\mathsf{E}_1}{\langle\, \mathsf{snd}\, x\,,\ \mathsf{fst}\, x\,\rangle : B \wedge A}\,\wedge\mathsf{I}}{\lambda x.\,\langle\, \mathsf{snd}\, x\,,\ \mathsf{fst}\, x\,\rangle : (A \wedge B) \supset (B \wedge A)}\,\supset\mathsf{I}^x$$

Inference Rules for Conjunction (Product Type)

Notation

Let Γ be a set of judgements and let $M : A$ be a judgement. The relation $\Gamma \vdash M : A$ means that there is a derivation with conclusion $M : A$ from the set of hypotheses Γ.

Inference Rules for Conjunction (Product Type)

Example

(i) A proof that $A \wedge B\,\mathrm{true} \vdash B \wedge A\,\mathrm{true}$.

$$\dfrac{\dfrac{A \wedge B\,\mathrm{true}}{B\,\mathrm{true}}\,\wedge\mathsf{E}_2 \qquad \dfrac{A \wedge B\,\mathrm{true}}{A\,\mathrm{true}}\,\wedge\mathsf{E}_1}{B \wedge A\,\mathrm{true}}\,\wedge\mathsf{I}$$

(ii) A proof that $M : A \wedge B \vdash \langle\, \mathsf{snd}\, M\,,\ \mathsf{fst}\, M\,\rangle : B \wedge A$.

$$\dfrac{\dfrac{M : A \wedge B}{\mathsf{snd}\, M : B}\,\wedge\mathsf{E}_2 \qquad \dfrac{M : A \wedge B}{\mathsf{fst}\, M : A}\,\wedge\mathsf{E}_1}{\langle\, \mathsf{snd}\, M\,,\ \mathsf{fst}\, M\,\rangle : B \wedge A}\,\wedge\mathsf{I}$$

Inference rules for Disjunction (Sum Type or Coproduct Type)

- Formation rule

$$\frac{A\,\mathrm{prop} \qquad B\,\mathrm{prop}}{A \vee B\,\mathrm{prop},}\ \vee\mathsf{F} \qquad \frac{A\,\mathrm{type} \qquad B\,\mathrm{type}}{A \vee B\,\mathrm{type}.}\ \vee\mathsf{F}$$

(continued on next slide)

Inference rules for Disjunction (Sum Type or Coproduct Type)

- Introduction rules

$$\frac{A\,\mathrm{true}}{A \vee B\,\mathrm{true},}\ \vee\mathsf{I}_1 \qquad \frac{M : A}{\mathsf{inl}\,M : A \vee B.}\ \vee\mathsf{I}_1$$

$$\frac{B\,\mathrm{true}}{A \vee B\,\mathrm{true},}\ \vee\mathsf{I}_2 \qquad \frac{N : B}{\mathsf{inr}\,N : A \vee B.}\ \vee\mathsf{I}_2$$

Inference rules for Disjunction (Sum Type or Coproduct Type)

- Introduction rules

$$\frac{A\ \mathrm{true}}{A \vee B\ \mathrm{true},}\ \vee\mathsf{I}_1 \qquad \frac{M : A}{\mathsf{inl}\, M : A \vee B.}\ \vee\mathsf{I}_1$$

$$\frac{B\ \mathrm{true}}{A \vee B\ \mathrm{true},}\ \vee\mathsf{I}_2 \qquad \frac{N : B}{\mathsf{inr}\, N : A \vee B.}\ \vee\mathsf{I}_2$$

- Elimination rule

$$\frac{A \vee B\ \mathrm{true} \quad \begin{matrix}\overline{A\ \mathrm{true}}^{\,u} \\ \vdots \\ C\ \mathrm{true}\end{matrix} \quad \begin{matrix}\overline{B\ \mathrm{true}}^{\,w} \\ \vdots \\ C\ \mathrm{true}\end{matrix}}{C\ \mathrm{true},}\ \vee\mathsf{E}^{u,w} \qquad \frac{M : A \vee B \quad \begin{matrix}\overline{u : A}^{\,u} \\ \vdots \\ N : C\end{matrix} \quad \begin{matrix}\overline{w : B}^{\,w} \\ \vdots \\ P : C\end{matrix}}{\mathsf{case}(M, u.N, w.P) : C.}\ \vee\mathsf{E}^{u,w}$$

Inference rules for Disjunction (Sum Type or Coproduct Type)

- Introduction rules

$$\frac{A\ \mathrm{true}}{A \vee B\ \mathrm{true},}\ \vee\mathsf{I}_1 \qquad \frac{M : A}{\mathsf{inl}\ M : A \vee B.}\ \vee\mathsf{I}_1$$

$$\frac{B\ \mathrm{true}}{A \vee B\ \mathrm{true},}\ \vee\mathsf{I}_2 \qquad \frac{N : B}{\mathsf{inr}\ N : A \vee B.}\ \vee\mathsf{I}_2$$

- Elimination rule

$$\frac{A \vee B\ \mathrm{true} \quad \begin{array}{c} \overline{A\ \mathrm{true}}^{\,u} \\ \vdots \\ C\ \mathrm{true} \end{array} \quad \begin{array}{c} \overline{B\ \mathrm{true}}^{\,w} \\ \vdots \\ C\ \mathrm{true} \end{array}}{C\ \mathrm{true},}\ \vee\mathsf{E}^{u,w} \qquad \frac{M : A \vee B \quad \begin{array}{c} \overline{u : A}^{\,u} \\ \vdots \\ N : C \end{array} \quad \begin{array}{c} \overline{w : B}^{\,w} \\ \vdots \\ P : C \end{array}}{\mathsf{case}(M, u.N, w.P) : C.}\ \vee\mathsf{E}^{u,w}$$

- Reduction rules

$$\mathsf{case}(\mathsf{inl}\ M, u.N, w.P) \Rightarrow_R N\,[\,u := M\,],$$

$$\mathsf{case}(\mathsf{inr}\ M, u.N, w.P) \Rightarrow_R P\,[\,w := M\,].$$

Inference rules for Disjunction (Sum Type or Coproduct Type)

Example

(i) A proof that $(A \vee B) \supset (B \vee A)$ true.

$$
\dfrac{
\dfrac{}{A \vee B \text{ true}}\,u
\qquad
\dfrac{\dfrac{}{A \text{ true}}\,v}{B \vee A \text{ true}}\,\vee\mathsf{I}_2
\qquad
\dfrac{\dfrac{}{B \text{ true}}\,w}{B \vee A \text{ true}}\,\vee\mathsf{I}_1
}{
\dfrac{B \vee A \text{ true}}{(A \vee B) \supset (B \vee A) \text{ true}}\,\supset\mathsf{I}^u
}\,\vee\mathsf{E}^{v,w}
$$

(ii) A proof that $\lambda u.\,\mathsf{case}(u, v.\,\mathsf{inr}\,v, w.\,\mathsf{inl}\,w) : (A \vee B) \supset (B \vee A)$.

$$
\dfrac{
\dfrac{}{u : A \vee B}\,u
\qquad
\dfrac{\dfrac{}{v : A}\,v}{\mathsf{inr}\,v : B \vee A}\,\vee\mathsf{I}_2
\qquad
\dfrac{\dfrac{}{w : B}\,w}{\mathsf{inl}\,w : B \vee A}\,\vee\mathsf{I}_1
}{
\dfrac{\mathsf{case}(u, v.\mathsf{inr}\,v, w.\mathsf{inl}\,w) : B \vee A}{\lambda u.\,\mathsf{case}(u, v.\,\mathsf{inr}\,v, w.\,\mathsf{inl}\,w) : (A \vee B) \supset (B \vee A)}\,\supset\mathsf{I}^u
}\,\vee\mathsf{E}^{v,w}
$$

Inference Rules for Truth (Unit Type)

- Formation rule

$$\frac{}{\top\ \mathrm{prop},}\ \top F \qquad \frac{}{\top\ \mathrm{type.}}\ \top F$$

Inference Rules for Truth (Unit Type)

- Formation rule

$$\frac{}{\top \; \mathrm{prop},}\;\top\mathsf{F} \qquad \frac{}{\top \; \mathrm{type}.}\;\top\mathsf{F}$$

- Introduction rule

$$\frac{}{\top \; \mathrm{true},}\;\top\mathsf{I} \qquad \frac{}{\langle\rangle : \top .}\;\top\mathsf{I}$$

Inference Rules for Truth (Unit Type)

- Formation rule

$$\frac{}{\top\ \mathrm{prop},}\ \top\mathsf{F} \qquad \frac{}{\top\ \mathrm{type}.}\ \top\mathsf{F}$$

- Introduction rule

$$\frac{}{\top\ \mathrm{true},}\ \top\mathsf{I} \qquad \frac{}{\langle\rangle : \top.}\ \top\mathsf{I}$$

- No elimination rule
- No reduction rule

Inference Rules for Falsehood (Empty Type)

- Formation rule

$$\frac{}{\perp\ \mathrm{prop},}\ \perp\mathsf{F} \qquad \frac{}{\perp\ \mathrm{type}.}\ \perp\mathsf{F}$$

Inference Rules for Falsehood (Empty Type)

- Formation rule

$$\frac{}{\bot\ \mathrm{prop},}\ \bot\mathsf{F} \qquad \frac{}{\bot\ \mathrm{type}.}\ \bot\mathsf{F}$$

- Elimination rule

$$\frac{\bot\ \mathrm{true}}{A\ \mathrm{true},}\ \bot\mathsf{E} \qquad \frac{M : \bot}{\mathsf{abort}\ M : A.}\ \bot\mathsf{E}$$

Inference Rules for Falsehood (Empty Type)

- Formation rule

$$\frac{}{\perp\,\mathrm{prop},}\ \perp\mathsf{F} \qquad \frac{}{\perp\,\mathrm{type}.}\ \perp\mathsf{F}$$

- Elimination rule

$$\frac{\perp\,\mathrm{true}}{A\,\mathrm{true},}\ \perp\mathsf{E} \qquad \frac{M : \perp}{\mathsf{abort}\,M : A.}\ \perp\mathsf{E}$$

- No introduction rule
- No reduction rule

Guiding Principles behind the Proposition-as-Types Principle

Guiding principles behind the proposition-as-types principle

Textbook (p. L4.6):

(i) *Every proposition corresponds to a type and vice versa.*
(ii) *Introduction rules correspond to value constructors.*
(iii) *Elimination rules correspond to value destructors.*
(iv) *For every deduction of* $A\,\mathrm{true}$ *there is a proof term* M *and deduction of* $M : A$*. We can effectively construct this top-down.*
(v) *For every deduction of* $M : A$ *there is a deduction of* $A\,\mathrm{true}$*. We can effective construct this bottom-up.*

Agda Implementation of Function Types (Implication)

The function type $\rightarrow$ is built-in in `Agda`.

Agda Implementation of Function Types (Implication)

Example

A proof and an implementation that $\lambda a.a : A \to A$.

$$\dfrac{\dfrac{}{a : A}\,a}{\lambda a.a : A \to A}\,\to\mathsf{I}^{a}$$

```
id₁ : {A : Set} → A → A
id₁ = λ a → a

id₂ : {A : Set} → A → A
id₂ a = a
```

Agda Implementation of Function Types (Implication)

Example

A proof and an implementation that $\lambda a.\lambda b.a : A \to (B \to A)$.

$$\dfrac{\dfrac{\dfrac{}{a : A}\,a}{\lambda b.a : B \to A}\,\to\mathsf{I}}{\lambda a.\lambda b.a : A \to (B \to A)}\,\to\mathsf{I}^{a}$$

```
ax1 : {A B : Set}→ A→ (B→ A)
ax1 = λ a _→ a

ax2 : {A B : Set}→ A→ (B→ A)
ax2 a _ = a
```

Agda Implementation of Product Types (Conjunction)

- Formation and introduction rules

$$\frac{A\ \text{type} \qquad B\ \text{type}}{A \times B\ \text{type},}\ \times\mathsf{F}$$

$$\frac{M : A \qquad N : B}{\langle M , N \rangle : A \times B.}\ \times\mathsf{I}$$

```
data _×_ (A B : Set) : Set where
  ⟨_,_⟩ : A→ B→ A × B
```

Agda Implementation of Product Types (Conjunction)

- Elimination and reduction rules

$$\frac{M : A \times B}{\text{fst}\, M : A,} \times\text{E}_1$$

$$\frac{M : A \times B}{\text{snd}\, M : B,} \times\text{E}_2$$

$$\text{fst}\,\langle\, M \,,\, N \,\rangle \Rightarrow_R M,$$
$$\text{snd}\,\langle\, M \,,\, N \,\rangle \Rightarrow_R N.$$

```
fst : {A B : Set}→ A × B→ A
fst ⟨ a , _ ⟩ = a

snd : {A B : Set}→ A × B→ B
snd ⟨ _ , b ⟩ = b
```

Agda Implementation of Product Types (Conjunction)

Example

A proof and an implementation that $\lambda x.\,\langle\,\mathsf{snd}\,x\,,\,\mathsf{fst}\,x\,\rangle : (A \times B) \to (B \times A)$.

$$\dfrac{\dfrac{\dfrac{\dfrac{}{x : A \times B}\,x}{\mathsf{snd}\,x : B}\,{\times}\mathsf{E}_2 \qquad \dfrac{\dfrac{}{x : A \times B}\,x}{\mathsf{fst}\,x : A}\,{\times}\mathsf{E}_1}{\langle\,\mathsf{snd}\,x\,,\,\mathsf{fst}\,x\,\rangle : B \times A}\,{\times}\mathsf{I}}{\lambda x.\,\langle\,\mathsf{snd}\,x\,,\,\mathsf{fst}\,x\,\rangle : (A \times B) \to (B \times A)}\,{\to}\mathsf{I}^{x}$$

```
×-comm₁ : {A B : Set}→ A× B→ B× A
×-comm₁ = λ x→ ⟨ snd x ,  fst x ⟩

 -- Using pattern-matching
×-comm₂ : {A B : Set}→ A× B→ B× A
×-comm₂ ⟨ a , b ⟩ = ⟨ b ,  a ⟩
```

Agda Implementation of Sum Types (Disjunction)

- Formation and introduction rules

$$\frac{A \text{ type} \qquad B \text{ type}}{A + B \text{ type},} +\mathsf{F}$$

$$\frac{M : A}{\mathsf{inl}\, M : A + B,} +\mathsf{I}_1$$

$$\frac{N : B}{\mathsf{inr}\, N : A + B.} +\mathsf{I}_2$$

```
data _+_ (A B : Set) : Set where
  inl : A→ A + B
  inr : B→ A + B
```

Agda Implementation of Sum Types (Disjunction)

- Elimination and reduction rules

$$\frac{M : A + B \qquad \begin{array}{c}\overline{u : A}^{\,u}\\ \vdots \\ N : C\end{array} \qquad \begin{array}{c}\overline{w : B}^{\,w}\\ \vdots \\ P : C\end{array}}{\mathsf{case}(M, u.N, w.P) : C,}\;{+}\mathsf{E}^{u,w}$$

$$\mathsf{case}(\mathsf{inl}\, M, u.N, w.P) \Rightarrow_R N\,[\,u := M\,],$$
$$\mathsf{case}(\mathsf{inr}\, M, u.N, w.P) \Rightarrow_R P\,[\,w := M\,].$$

```
case : {A B C : Set}→
       A + B→
       (A→ C)→
       (B→ C)→
       C
case (inl a) f g = f a
case (inr b) f g = g b
```

Agda Implementation of Sum Types (Disjunction)

Example

A proof and an implementation that $\lambda u.\, \mathsf{case}(x, a.\, \mathsf{inr}\, a, b.\, \mathsf{inl}\, b) : (A + B) \to (B + A)$.

$$
\dfrac{
\dfrac{
\dfrac{}{u : A + B}\,x
\qquad
\dfrac{\dfrac{}{a : A}\,a}{\mathsf{inr}\, v : B + A}\,{+}\mathsf{I}_2
\qquad
\dfrac{\dfrac{}{b : B}\,b}{\mathsf{inl}\, b : B + A}\,{+}\mathsf{I}_1
}{\mathsf{case}(x, a.\mathsf{inr}\, a, b.\mathsf{inl}\, b) : B + A}\,{+}\mathsf{E}^{a,b}
}{\lambda x.\, \mathsf{case}(x, a.\, \mathsf{inr}\, a, b.\, \mathsf{inl}\, b) : (A + B) \to (B + A)}\,{\to}\mathsf{I}^{x}
$$

```
+-comm₁ : {A B : Set}→ A + B→ B + A
+-comm₁ = λ x→ case x (λ a→ inr a) (λ b→ inl b)

-- Using pattern-matching
+-comm₂ : {A B : Set}→ A + B→ B + A
+-comm₂ (inl a) = inr a
+-comm₂ (inr b) = inl b
```

Agda Implementation of Unit Type (Truth)

- Formation and introduction rules

$$\frac{}{\top\ \mathrm{type},}\ \top\mathsf{F}$$

$$\frac{}{\langle\rangle : \top.}\ \top\mathsf{I}$$

```
data ⊤ : Set where
  ⟨⟩ : ⊤
```

Agda Implementation of Empty Type (Falsehood)

- Formation rule

$$\frac{}{\bot\ \text{type.}}\ \bot\text{F}$$

```
data ⊥ : Set where
```

Agda Implementation of Empty Type (Falsehood)

- Formation rule

$$\frac{}{\bot\ \mathrm{type.}}\ \bot\mathsf{F}$$

```
data ⊥ : Set where
```

- Elimination rule

$$\frac{M : \bot}{\mathsf{abort}\ M : A.}\ \bot\mathsf{E}$$

```
abort : {A : Set} → ⊥ → A
abort ()
```

Summary of The Proposition-as-Types Principle

Summary of the proposition-as-types principle

Textbook (pp. L4.11–12):

(i) Propositions as types

(ii) Proofs as programs

(iii) Proof reduction as computation

References

Frank Pfenning (26th Jan. 2023). Lecture Notes on Proofs as Programs. Material for the course 15-317 Constructive Logic at Carnegie Mellon University, Spring 2023. URL: https://www.cs.cmu.edu/~fp/courses/15317-s23/ (cit. on p. 2).

Morten-Heine Sørensen and Paul Urzyczyn (2006). Lectures on the Curry-Howard Isomorphism. Vol. 149. Studies in Logic and the Foundations of Mathematics. Elsevier (cit. on p. 12).